

MAE 321 Wing Project

Final Design Report

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Abstract

In this design report the aim is to lay out the design for, theory behind, manufacturing process of, and stress test results of, an aircraft wing, which should be able to not inelastically yield when subject to 95.5 lbf when set up as a cantilever, all while following the given constraints. Aluminum 7075-T6 was used for the beam (due to its high yield stress limit), 3D printed PLA (due to its light weight) for bulkheads (except for leading and ending bulkheads which were of aluminum 6061) as well as leading edge, 7075-T6 for the top of the airfoil surface (high tensile strength), 2024-T3 for the bottom surface (lighter, and higher compressive strength), and 6061 for fasteners. The beam was manufactured with a hole pattern to reduce weight, as well as a solid bridge in the middle of the beam to directly alleviate stresses that were concentrated there based on simulated results. A quantity of nine bulkheads were used. Five rivets were used on the fixed bulkhead, three on the free one, and one on each of the seven others. The equipment necessary for manufacturing included a CNC 3-Axis milling machine, a water jet cutter, and an SLS 3D printer. The beam was tested by being fixed via the largest bulkhead, while being pulled by a winch hanging from the other end. The force was increased incrementally, up to 385 lb, as we had to stop increasing the weight used out of safety. The beam, and by extent, the wing, did not fail, though there was visible damage, most notably with all five rivets sheared off on the fixed end, the buckling of the skin on the lower portion of the airfoil, and with the leading edge cracking along the 3D print layer lines. The detachment of the rivets caused an unwanted distribution of force, which weakened the structure. We also noted that the deflection increased linearly with force. The minimum specified weight of 95.5 lb resulted in 3.625 inch deflection, and went up to 19.5 inches for the 385 lb weight.

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Introduction

I. Problem Description

The wings of aircrafts are designed, tested and built to withstand immense forces caused by lift and turbulence during flights. They are engineered to flex under the action of these forces and absorb and dissipate energy without deforming permanently, while also being light-weight, aerodynamic and corrosion-resistant to allow for flight at very high altitudes. These requirements are met through the use of specialized metals/alloys, structural designs and manufacturing process, accompanied by rigorous testing. High-strength, lightweight materials such as Aluminium alloys are often used to handle the enormous forces without significantly weighing down the aircraft. As the wings are only fixed to the main body of the plane on one end and lack any other supports, they are often modelled as cantilevers with complex, optimized and well-tested designs for internal spars and ribs to enhance support while minimizing weight.

Through this project, we specifically aim to study wing designs that maximize load withstood without significant deformation while minimizing weight – disregarding aspects like aerodynamic performance and resistance to corrosion that more general aircraft wing design should otherwise consider. This involves choosing materials for, researching, designing and stress testing the central I-beam, bulkheads, airfoil surface and rivets to construct a 2-foot long model for an aircraft wing.

The stresses experienced by a wing during flight vary continuously depending on velocity of the aircraft, turbulence experienced, and local wind speed and direction. We will simplify the forces that typically act on a wing into a singular 95.5 lbf force acting on the smaller side of the model, 23 inches away from the point at which the wing is held horizontally. As this point is furthest from the plane's body, a force acting here will cause the greatest torque and thereby exhibit the greatest likelihood of deformation under stress. We will stress test our designs using PTC CREO to identify optimal designs that satisfy all our dimensional criteria and then construct the final model using CNC and other machining techniques. Finally, we will physically test our wing design to investigate if it is able to meet our design requirements while accomplishing the intended purpose of withstanding the 95.5 lbf force.

II. Objectives and Criteria

The objective of this project is to design and manufacture a wing from a predetermined airfoil that is able to support a weight of 95.5 lbs at the smaller end of the wing. We chose to focus closely on four specific aspects of this project. As explained above, it is important that the wing exhibits sufficient structural integrity to successfully support the applied load without failure. Likewise, minimizing deflection under loading and weight of the wing are also crucial goals to pay attention to during design. Lastly, when constructing the wing, it is important that we machine it with care and attention -to-detail (i.e. good workmanship) to ensure the physical model is identical to the Creo design on which stress testing was performed.

The aerodynamics of the wing and the external features of the wing were predetermined and kept unchanged throughout the wing's manufacturing process. As explained earlier, the primary focus of this project is to maximize strength and minimize weight and deflection. Therefore, aerodynamic suitability and resistance to corrosion were disregarded. We want to minimize the weight of the wing to increase the payload the airplane will be able to carry, and by removing unnecessary weight from the wing we allow for a greater percentage of the plane's takeoff weight to be dedicated to cargo and passengers. Similarly, the deflection is minimized to increase the effective lift on the wing and decrease the risk of permanent deformation in its internal structure. Good workmanship is important as it can affect the actual strength of

the wing. Parts containing cracks or improperly made rivets would decrease the total weight the wing is able to support, thereby reducing the capability of our model wing.

The machines and tools used to build the wing will be restricted to those available in the MAE department's machine shop. All steps of the model's construction must be completed using only 3D printable materials and aluminum alloys available from McMaster-Carr Supply company. We will have access to two 3D printers: an FDM and a nylon SLS.

Background Theory

I. Project Specifications

The wing for this project must be constructed using the supplied profile of the airfoil, but can be rescaled based on our chosen design parameters. All sheets and beams must be made of aluminum of any grade available on McMaster-Carr. Any number of bulkheads may be used and made from 3D-printed PLA, Nylon, or water-jetted aluminum, however, the first is required to be aluminum.

The wing is required to be 22" long with a central beam extending 1 ½" past the wing end, resulting in a full beam length of 23 ½". The end of the beam is required to have a 3/16" hole drilled horizontally 23" from the fixed end of the wing to facilitate loading. The fixed end of the beam is required to extend 2 inches past the fixed end of the wing with a ¾" inch square side lengths and a ¼ inch hole drilled vertically 1 inch from the fixed end. Additionally, the bottom surface area of the wing must be between 98-102 square inches.

The first aluminum bulkhead is required to have a ¾" square hole for the beam as well as 10-32 threaded holes on a 1" hole pattern. The wing needs to be secured to the fixture using at least three 10-32 threaded holes spaced 1 inch apart as the beam extension is secured with a ¼-20 grade 8 screw.

Grading is based on several wing performance factors including: 1) Holding the specified weight of 95.5 lbs, 2) Minimizing weight, 3) Minimizing deflection, 4) Ensuring the 98-102 sq. in. bottom side surface area, and 5) Workmanship.

II. Stress and Strain

There are two central concepts that are important to clarify in the theoretical portion of this paper. The first is stress, which is a measurement of the force per unit area applied on a surface. Since stress is a vector quantity, it can be separated into two components, namely the normal stress and shear stress. The normal stress, $\sigma = \frac{F}{A}$ is the stress component in the direction normal to the surface. The shear stress, defined $\tau = \frac{F}{A}$, where F is the applied force and A is the surface area, is the stress applied parallel to the surface. When either the normal stress or the shear stress exceeds a certain threshold (σ_{yield} and τ_{yield} respectively), the material fails. Thus, to prevent a component from failing, it is important to design them by increasing the yield strength, decreasing the force applied on them, or increasing the area on which the force is applied.

The second concept is strain, which is a dimensionless quantity describing the deformation of a material. Like stress, it can also be separated into components called the normal strain and shear strain. Normal strain is the ratio of the change in length to a reference length. It can be calculated in two ways. The first is the engineering strain ($e = \frac{\Delta L}{L}$), where ΔL is the change in length of the material and L is the original length of the material. This is only an accurate measurement of the relative deformation for small deformations since it is only under this regime where $e \propto \Delta L$. This stems from the fact that it is a

derivation of Hooke's law ($F_s = kx$), which is an empirical first-order approximation. The true strain on the other hand is defined as $\epsilon = \ln(\frac{\Delta L}{L})$, and is more accurate than the engineering strain. The shear strain is similar, except that it measures the relative angular deformation of an object.

III. Buckling

Another very important concept when structures are under compressive loads is buckling. This is a sudden deformation that occurs when a structure is under compression exceeding a critical load value. Euler's Buckling Formula ($P_{cr} = \frac{\pi^2 EI}{L_e^2}$) shows that this critical load value is dependent on the Young's Modulus of the material (E), the effective length (L_e), and the second moment of inertia (I). The effective length of a structural member is dependent on its constraints. For example, beams that are fixed on both ends will buckle at different critical loads than beams that are fixed at one end and free at the other end. When designing this wing, the bottom surface will be under compression, so avoiding this phenomenon will be an important consideration to avoid sudden failure.

Design Theoretical Analysis

I. Central Beam

When designing the central beam, the first item we deliberated was which material would be best suited for use. We decided between aluminum 6061-T6, 7075-T6, and 2024-T3. The decision to not use 6061, nor 2024-T3 was based on their comparatively poor yield stress range, especially that of 6061. Comparing Fig. I.1 and Fig. I.2 shows that even at the minimum force specification of 92 lbf, the stress of about 60 ksi was high compared to the yield strength range being 34.8 to 40.6 ksi for 6061-T6, whereas even with an added 8 lbf, the mid 60 ksi reading on the 7075-T6 was more safely within its range of 52.1 and 76.9 ksi. This information regarding max yield strength was found via the CES Edupack database software. These two figures served as a control, as they were the most unmodified versions of the beam that we simulated.

After the material of 7075-T6 was chosen, we introduced a hole pattern which was meant to mimic the shape of triangular trusses such that the material left over would allow force to disperse in diagonal paths reminiscent of more traditional trusses. This pattern is a simplification of an earlier idea which was to create circles of relatively large diameter and leave a line of material to go through it diagonally, but after including the fillets of 1/16 inch radius which would show the shape that would actually result from the nature of the machining process the pattern looked like an array of bean-shaped holes, and as their size decreased along with the taper of the beam, it became impossible to add the fillet (let alone having clearance for the machining bit to enter), which is why we simplified the entirety of the bean array to holes, as is shown in Fig. I.4. This design, although lighter than the control, picked up relatively high stresses. These higher stresses were concentrated near the center beam. This location is interesting because the pattern for the holes was such that one half-length of the beam had holes of a particular size, and the half that was more tapered had smaller diameter holes. The location of abrupt change in hole size was, whether coincidentally or not, at or near the area of highest recorded stress.

We tried to solve this problem by introducing a bridge at that middle section of the beam, and as shown in Fig. I.5, it helped relieve the stress only locally, as high (although lower) stresses were depicted at either end of the bridge. For the final design of the beam, as shown in Fig. I.6, we included the 1/16 inch radius rounds which would result from the machining process, which seemed to result in relatively lower

stress readings throughout, comparable to the control beam. It is also worth noting that, out of curiosity of the effect of the fillets on the stress readings, we tested a beam, shown in Fig. I.3, that was essentially the control with 0.2 in rounds in place of what would be 1/16 in radius rounds, and the readings were more favorable than any of the other designs, but it would be relatively heavy (for summary, see Table A1).

II. Bulkheads

In designing the bulkheads the goal was to optimize the number of bulkheads, material and geometry to minimize the weight while maintaining necessary load transfer and structural integrity.

In determining the materials to use for the bulkheads, PLA, Nylon and Water-jetted aluminum were considered. PLA and Nylon are both able to be 3D printed and are lightweight compared to the water jetted aluminum. PLA has a density of 0.045 lb/in³ and Nylon has a density of 0.043 lb/in³. This makes them more suitable as the material for the intermediate bulkheads where the structural demands are lower. Aluminum with a higher density of 0.0975 lb/in³ is therefore a better choice for the higher stress locations. To ensure effective load transfer at the end of the wing, we choose aluminum as the material for the first and last bulkheads where the stresses from the beam and external load will be the greatest. The bulkheads in between will be constructed using 3D printed PLA to provide support while minimizing weight.

To determine the number of bulkheads we ran static and buckling analysis in creo on the simulated wing. The buckling analysis was run under a constant 90 lb load with varying number of evenly spaced bulkheads ranging from 5 - 9. From these simulations we looked at the Beam Load Factor (BLF) which is a measurement of structural performance under bending. The BLF values were: 0.802 for 5 bulkheads, 0.722 for 6 bulkheads, -1.818 for 7 bulkheads, 0.923 for 8 bulkheads and -1.315 for 9 bulkheads. Due to these results, we considered using either 8 or 9 bulkheads but decided to use 9 bulkheads in our final design. The 9 bulkheads over 8 will increase the number of support points for the skin and reduce the risk of local buckling. The added constant along the beam will also help distribute the load applied more evenly therefore reducing the stress concentrations and potential for deformation. Though having 9 bulkheads will increase the weight we determined it would be a marginal increase due to PLA being a lightweight material and therefore was acceptable.

III. Airfoil Surface

When designing the skin of our airfoil, our decisions were guided by two main constraints: the aluminum material requirement and the offerings from McMaster-Carr's selection. After reviewing McMaster's offerings, we identified the two best aluminum alloy candidates to be 2024-T3 and 7075-T6 based on their material properties. In comparison, the other widely used alternative, 6061 alloys, were worse in nearly all relevant categories for this application.¹

As the load is applied to the end of the wing, the structure will experience bending, and according to classic beam theory, the top surface will stretch and the bottom compresses, resulting in tensile stresses within the top skin and compressive stresses in the bottom skin. Consequently, when analyzing material offerings, we focused on the tensile and compressive characteristics of materials to guide our choices and determine which would best withstand the given load.

¹Aerospace Specification Metals Inc., "ASM Material Data Sheet: 6061-T6 Aluminum," <https://asm.matweb.com/search/specifcmaterial.asp?bassnum=ma6061t6>.

Analyzing the material properties via ASM Aerospace Specification Metals Inc., we found that 7075-T6 offers higher tensile strength (83 ksi) and lower compressive modulus (10,608 ksi),² while 2024-T3 boasted lower tensile strength (70 ksi) and high compressive modulus (10,812 ksi).³ It is worth noting that the compressive modulus is not directly equivalent to compressive strength, however, these figures align with conventional literature on material selections for upper and lower wing covers of the 7XXX plate alloys and 2X24 alloys, respectively.⁴

For the top skin, 7075-T6 was selected because of its higher tensile strength of 83 ksi and yield strength of 73 ksi, which provide strong resistance to tensile forces which would be exerted on the upper skin by the 95.5 lb load. We observed that 7075-T6 had a slightly higher density than 2024 (0.102 lb/in³ vs 0.1 lb/in³), however, this difference is nominal and we consider it negligible given the tensile improvements. For the bottom skin, 2024-T3 was selected because of its superior compressive strength and toughness characteristics. Additionally, since the lower skin is not under tensile loads, it is advantageous to choose 2024-T3 for its lower density, given its lower tensile strength requirements.

For choosing the thickness measurements we simulated static stress analyses using CREO, varying the top/bottom thickness based on availability of McMaster-Carr stock (Appendix, Table A2). After testing a combination of McMaster thinnest stocks, we noted maximum approximate stresses on the top skin of around 15 KSI with the combination of 0.032" thickness of the top skin and 0.02" on the bottom (see Figs. III.1–4). This was approximately 25% of the yield stress of 61 ksi.⁵ The stresses on the bottom skin were less significant, resulting in the decision to choose a thinner metal and reduce the wing's overall weight. This incorporates an appropriate safety factor given the future additions of rivets and other component modifications which were not part of the simulation and would likely increase the maximum stress to some degree.

By selecting these two distinct metals and tailoring their thicknesses to withstand their respective loading conditions, the airfoil skin is optimized for both tension and compression while maintaining weight efficiency and relatively simple structural design.

IV. Component Fasteners

While in the above analyses the various components of the simulated wing were bonded together at their interfaces, the final real-world assembly will require the use of fasteners to maintain the connections between them. Due to the difficulty of accessing the interior of the wing to secure many of these, aluminum pop rivets were selected for this task, with a given diameter of 1/8". Notably, these rivets are manufactured out of a weaker aluminum alloy, and thus can only withstand around 20 ksi of stress in the surrounding material before risking failure.⁶

The first location chosen for rivet analysis were those that secure the airfoil surface to the bulkheads within the wing. While each bulkhead will be riveted to the skin of the wing, the two

² Aerospace Specification Metals Inc., "ASM Material Data Sheet: 7075-T6 Aluminum," <https://asm.matweb.com/search/specifimaterial.asp?bassnum=ma7075t6>.

³ Aerospace Specification Metals Inc., "ASM Material Data Sheet: 2024-T3 Aluminum," <https://asm.matweb.com/search/specifimaterial.asp?bassnum=ma2024t3>.

⁴ Prasad, N. E., A. A. Gokhale, and R. J. H. Wanhill. 2014. *Aluminum-Lithium Alloys: Processing, Properties, and Applications*. Elsevier.

⁵ McMaster-Carr, "High-Strength 7075 Aluminum Sheets and Bars," <https://www.mcmaster.com/products/aluminum-alloy-7075/high-strength-7075-aluminum-sheets-and-bars-7/>.

⁶ Per discussion with Glenn during the lab period. Many other ideas in this section were clarified in this manner as well, which were quite useful in making fastener design decisions.

aluminum bulkheads on the ends of the wing are primarily responsible for carrying this aspect of the load applied, and thus initial analysis was focused on these connections. To examine the stresses around the rivets, we created an assembly in CREO consisting of two overlapping rectangular plates with a variable number of evenly-spaced $\frac{1}{8}$ " holes through where they sit atop one another (see Fig. IV.1). Using CREO's analysis features, these holes were modeled as being held together with 6061 aluminum fasteners instead of material bonding between the two 7075 aluminum components (chosen due to the anticipated higher tension load on the upper surface of the wing). To test the load on this connection, simulations were run by holding one plate fixed and applying a 600-lbf tensile load to the surface of the other. This load was selected during preliminary testing in order to mirror expected stresses in the skin while slightly overcompensating to allow for a safety factor. The results of these analyses for 4, 5, and 6 rivets are displayed in Figs. IV.2–4. As can be seen, the 5-rivet design produced regions of around 17 ksi stress, which was a substantial improvement over the 24 ksi regions in the 4-rivet design, while notably the improvement to 16 ksi regions in the 6-rivet design was less major. Since 5 rivets were sufficient to reduce the stresses below failure risk while allowing for a greater ease of practical installation, we chose to use 5 rivets to secure the skin at these vital locations. Due to the hand-drilled nature of the rivet connections, no inquiry was put into the spacing of the rivets, as this would be practically difficult to control during assembly for minimal (if any) improvements in performance.

Outside of this primary rivet location, the rest of the skin must also be fastened to the intermediate bulkheads. The primary function of this is to minimize buckling of the skin, as by securing the lower surface along each bulkhead the beam is only able to buckle in the shorter length between each fastening (rather than the much easier buckling mode across the entire surface); however, only a single connection is really required to achieve this stability. This is increased by the hollow design of the intermediate bulkheads, which will allow the end of the rivet to more completely deform over the interior surface and provide resistance against normal stresses. Since these connections will not be carrying virtually any of the lateral load of the structure, as this is being accomplished by the rivets on the ends, this single rivet is thus sufficient, and hence we chose to use a single rivet for each intermediate bulkhead.

The final connections to consider in the design are those between the bulkheads and the beam and the bulkheads and the leading edge. Beginning with the connections to the beam, due to spatial constraints it is only reasonable to fit a single rivet for each connection, offset to the side as to avoid disrupting the stability of the web. As the upper side of the beam would be under tension, rivet holes drilled here would be stretched under deformations, leading to instabilities in the position of the connection; thus, we chose to place the connections on the underside to avoid this phenomenon. As for the leading edge, minimal load will be carried here, and thus connections were placed on the bottom for primarily aesthetic reasons.

V. Additional Practical Operation Tests

All of the above analysis was performed on a static wing, which, while good for ensuring the wing can carry the load that it needs to, does not provide complete information on how the wing would perform if actually put into operation as part of a larger aircraft. In this regime, two additional factors become important: the effects of repeated loading and unloading caused by repeated use, and the impact of vibrations from the motion of the air or a motor elsewhere in the craft.

Materials tend to lose some of their strength when subjected to repeated loading and unloading in a phenomenon called fatigue, which could lead to failure at much lower loads than initially designed for.

Using a CREO simulation on the beam as the primary structural component,⁷ we ran a fatigue analysis for 20 load-unload cycles, the results of which can be seen in Fig. V.1. As can be seen, while most of the beam would last the full 20 cycles, some portions of the upper flange would fail after repeated use, with the part lasting around 5 cycles before failure. While this is suitable for our own testing purposes, further considerations would need to be taken into account when designing a fully operational wing.

In addition, vibrations through a structure may lead to resonant phenomena in which the structure amplifies the vibrations received, which may cause significant additional stresses and a sudden failure far below the designed load. To determine the modal frequency at which such a resonant phenomenon might occur for our wing, we ran a modal analysis on our CREO simulated beam, being the primary structural component that would be involved in resisting a lateral vibrational mode. Upon its completion, the simulation predicted a vibrational mode around 42 Hz; as such, if this wing were to be put into operation, frequencies around 42 Hz and its harmonics should be avoided to minimize the impact of resonance.

Procedure

I. Manufacturing

Following the submission and approval of the preliminary design, the wing component was fabricated in the MAE shop to test the real-world outcomes of the major structural design decisions.

Manufacturing commenced by machining the primary load-bearing beam from a 1" x 1" bar of AL 7075 stock. This utilized a two-step operation on the CNC 3-Axis milling machine, which cut each side of the beam separately. This utilized a specialized mount that secured the beam to the base of the CNC while the machine was in operation.⁸ During the second operation, custom spacers were used to allow for compression to the base while minimizing warping strains, and shims will be used to reduce vibration in the free space between these.

Structural bulkheads were also produced using two distinct approaches. For each, the provided standard airfoil profile was used and integrated into the following processes. For the aluminum bulkheads at the base and tip of the wing, a water jet cutter was used before tapping the primary bulkhead's mounting holes with 10-32 threading. The remaining seven bulkheads were produced via SLS additive manufacturing, utilizing PLA material with 10% infill, pattern grid, 0.2 mm layer height, and a 3-layer thick shell. The leading edge was also printed in PLA material and the same print settings in three sections, divided equally along the length of the wing.

The assembly of the wing continued by sliding the bulkheads into their prescribed positions along the beam for a snug, contact fit, and then secured using a single underside rivet. The three pieces of the leading edge were attached in sequence with rivets into each bulkhead, also on the underside of the wing.

The wing's skin was cut from flat sheets ordered from McMaster-Carr by placing them against the beam/bulkhead assembly and tracing the outline needed to cover each side. After the profiles were traced, the band saw was used to make rough cuts, which were then smoothed and deburred. The skin's outer surface was cleaned and polished with fine-grit sandpaper for visual satisfaction, and each piece was

⁷ Both of these analyses were run on the beam only, as this is the primary structural component of the wing itself. However, other portions of the wing design (such as the skin) are likely to provide important changes in some of these properties, and further testing of the full design is thus needed to more completely characterize the impact of these phenomena on our design.

⁸ Glenn and Al have prepared a template to be used for this mount involving screwing holes through the stock and into a larger base plate in CREO, which we used. This template also allowed for a 2-inch extension of the beam on the larger end of the beam, to be used in mounting the wing to the testing apparatus.

bent using an apparatus in the shop to approximate the curvature of the airfoil on the ends to allow for easier installation.

The wing's underside 2024-T3 skin was attached to the assembly with three rivets on the tip end bulkhead in hand-drilled rivet patterns as previously specified. Additionally, four rivets were fixed along the primary base bulkhead, and one was placed at approximately the center of each bulkhead along the length of the wing. Rivet length was adjusted to correspond to the depth of the material each was attached to. The upper 7075-T6 skin was attached similarly, with three on the tip bulkhead, five on the primary, and one on each of the other seven bulkheads. This completed the manufacturing of the wing, which next underwent testing.

II. Testing

Testing Apparatus

The first step to testing the wing was to mount the wing such that its base was held fixed, as in our theoretical simulations. To do this, three 10-32 screws were fastened through the holes in the base bulkhead, and a vertical hole was hand-drilled through the beam extension to pass a vertical bolt through. This mount was then affixed to the back of one of the milling machines in the shop using 4 more heavy-duty screws, and a bar was C-clamped to the upper part of this mount to provide a motionless basis from which the deflection of the wing could be measured.

To load the wing, a 1/4"-20 screw was put through the hole in the beam extension at the wingtip which was connected via a pulley to a winch which began on the ground. Using a simple motor and switch, this winch could be raised and lowered such that the load could be prepared without weighing on the wing and then it could be applied, avoiding potentially overloading the wing with instabilities caused by loading irregularities. The load itself comprised of at a minimum the winch-pulley system (weighing about 5.5 pounds) and a steel plate (weighing 65 pounds), upon which additional loads could be added using either 25-pound lathe chucks or additional 65-pound steel plates. To hold the chucks in place atop the plates, duct tape was used. Fig. VI.1 shows the wing completely fastened to the apparatus in its initially unloaded state.

Testing to original specifications

Once the beam was placed in the apparatus, we measured the initial distance between the reference bar and the top surface of the very tip of the beam using a measuring tape to serve as a baseline value by which we could calculate the net deflection for each load. After this, the wing was subjected to the minimum trial load of only the base steel plate (65 pounds) by powering the winch until the steel plate left the ground, after which the distance between the reference bar and the wingtip was again measured, and the net deflection was calculated and recorded. The winch was then lowered back to the ground, and a chuck was placed onto the plate to achieve the specified 95-pound load. The testing procedure was then repeated: raising the load off of the ground, measuring the net deflection, and then lowering the load back to the ground. Fig. VI.2 shows the wing loaded at this specified weight.

Testing beyond specifications

After testing the wing to the required 95 pounds, we decided to keep the wing secured to the testing apparatus and to continue testing beyond the required specifications until a major mechanical failure occurred. A major mechanical failure in this context would be an event that caused a sudden large change in the deflection. The weight was increased by adding either a 25 pound lathe chucks or 65 pound

steel disks to the winch, creating the following overall loads (in pounds): 130, 155, 180, 205, 230, 255, 280, 285, 310, 335, 360, and 385. For each of these loads, the requisite weights were applied on the ground, then raised off of the ground with the winch, where the net deflection was measured, after which the winch was lowered back to the ground. Notably, no failure ended up occurring, as while the goal was to increase the load until the wing failed, we decided to halt testing before this point due to an increasing concern that the testing apparatus could be damaged by the extreme load.

Conclusion

I. Results

For each weight we applied to the end of the wing, the wing did not fail, and we stopped out of safety at 385 pounds. With each increasing weight, the deflection also increased in a generally linear manner, save for a few outliers, as is shown in Fig. VII.6. We began with a load of 65 pounds resulting in a deflection of 2.5 inches and ended with a load of 385 pounds resulting in a total deflection of 19.5 inches (Table VII.1). The required weight of 95.5 pounds resulted in a 3.625 inch deflection (Table VII.1). Table VII.1 also includes the changes in deflection between each added weight. The largest change was between 180 pounds and 205 pounds with a change of 4.5 inches, and the smallest was zero change between 280 pounds and 285 pounds. However, this was only the result of a 5 pound difference. For all the other 25 pound differences, the smallest change in deflection was 0.5 inches between 285 pounds and 310 pounds. There was also a negative change in deflection of 0.5 inches between 205 pounds and 230 pounds. Finally, with the addition of 155 pounds, there was no recorded deflection, but the wing did successfully hold this load.

While there was no significant failure at any weight tested, there was damage to multiple sections of the wing. The skin had warping and buckling between rivets (Figs. VII.1 and VII.5) and multiple popped rivets on the first bulkhead (Figs. VII.3 and VII.4). There were fractures and cracks along the leading edge layer lines (Figs. VII.1 and VII.2). The tape along the trailing edge was stretched and torn (Fig. VII.3), and there was delamination of the bulkheads (Fig. VII.5) and deformation of the beam (Fig. VII.3).

II. Discussion

After stopping at 385 pounds of load, there were clear signs of damage. The first was the buckling of the skin, especially on the bottom side of the wing. At 385 pounds of load, the deflection was 19.5 inches (Table VII.1). In order for this deflection to occur, the bottom of the wing must compress, while the top must expand. This extreme compression is what caused the skin to buckle in between rivets. Should load have continued to be increased, this buckling would likely have become worse. Another point of failure was at the leading edge of the wing. Since the leading edge is 3D printed with layers, it is anisotropic, meaning that it is stronger in one direction than another. This is clearly displayed in the fact that it fractured along the layer lines, resulting in cracks propagating along the leading edge. If there was a way to have the leading edge strength be equal in all directions, it likely would have failed throughout the leading edge, rather than in a certain direction, and likely may have delayed its fracturing.

The main failure point, however, were the rivets used in the wing. Because the rivets are relatively small and fasten much of the wing together, they experience significant shear load and have

very little cross sectional area to dissipate these forces (see Background Theory). All five of the pop rivets attached to the first bulkhead sheared due to the applied load on the wing. In most cases, the failure of the rivets results in the complete failure of the wing, since the same load experienced by the wing is no longer dissipated over these five rivets, and so must be supported by the remaining rivets, which are equally weak.

III. Implications for Future Work

While the wing withstood all the applied loads without catastrophic failure, there are several areas that could potentially be refined based on the experimental outcomes and observed damage patterns. The areas with repeated failure were around the rivet, including popped rivets, stretched rivet holes, and skin buckling. This implies that more detailed analysis could be done for rivet distribution, strength, and the capability of the material. In the future the spacing of the rivet should be optimized and potentially alternative fastener types or reinforcement strategies to reduce local stress concentrations. Also, since the final design appeared to be significantly over-engineered, reducing the overall weight by strategically lightening the beam without compromising structural integrity could be explored. This could involve topological optimization or iterative finite element analysis to identify low-stress regions that could be removed or thinned.

Future testing could also expand to include long-term fatigue and modal analyses of the complete wing structure, rather than just the beam. Our current simulations indicated failure due to fatigue would begin after as few as five cycles in certain regions, and a modal resonance near 42 Hz. Therefore, future studies could include full-system simulations, long-term fatigue tests, and potentially wind tunnel experiments to evaluate how the wing responds to aerodynamic forces during real-world operation.

Appendix

I. Central Beam

Diagram	Material	Fillet (in)	Hole Pattern	Bridge	Load (lbf)	Notes
I.1	7075-T6	No	No	No	100	[Control] Partially successful
I.2	6061-T6	No	No	No	95	[Control] Partially successful
I.3	7075-T6	R 0.2	No	No	95	Successful
I.4	7075-T6	R 1/16	Yes	No	95	Failed
I.5	7075-T6	No	Yes	Yes	95	Partially Successful
I.6	7075-T6	R 1/16	Yes	Yes	95	[Final] Successful

Table A1 - CREO Static Stress Analysis for I-beam Designs



Fig. I.1: AL 7075-T6 Control I-beam with original design (no holes, no bridge, no fillet) with 100 lbf force acting on end.

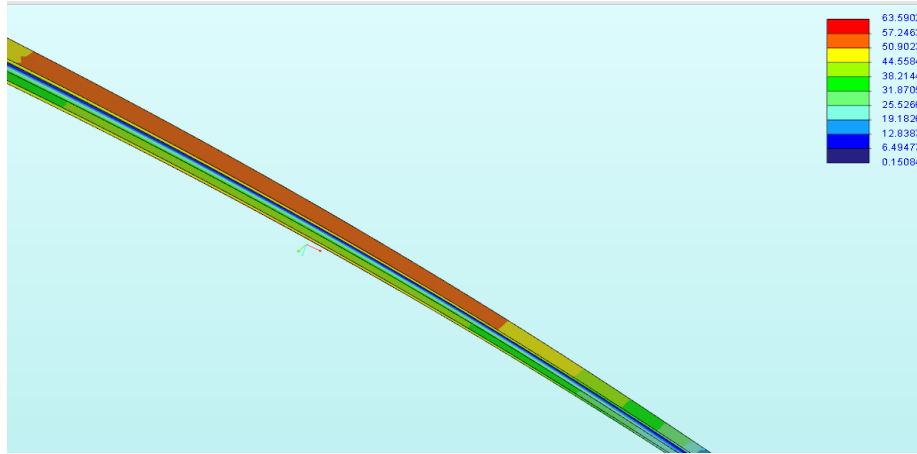


Fig. I.2: AL 6061-T6 Control I-beam with original design (no holes, no bridge, no fillet) with 95 lbf force acting on end.

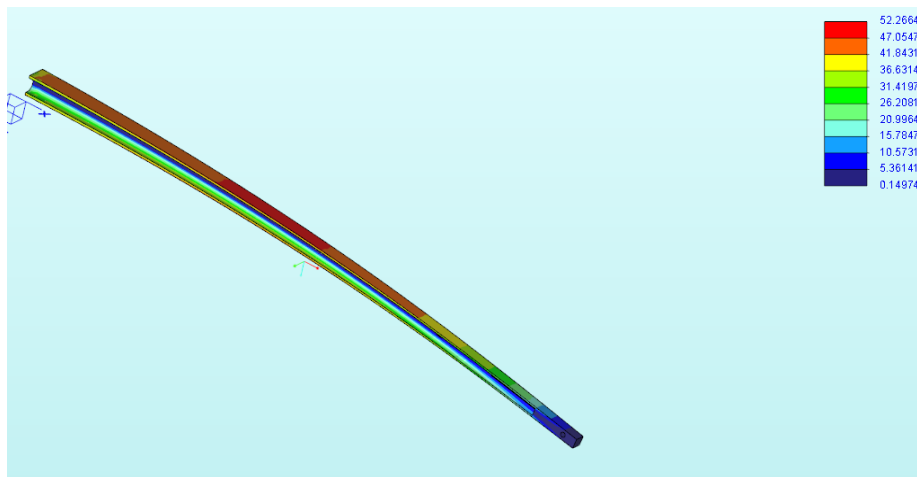


Fig. I.3: AL 7075-T6 Modified I-beam with 0.2 in radius fillet (no holes, no bridge) with 95 lbf force acting on end.

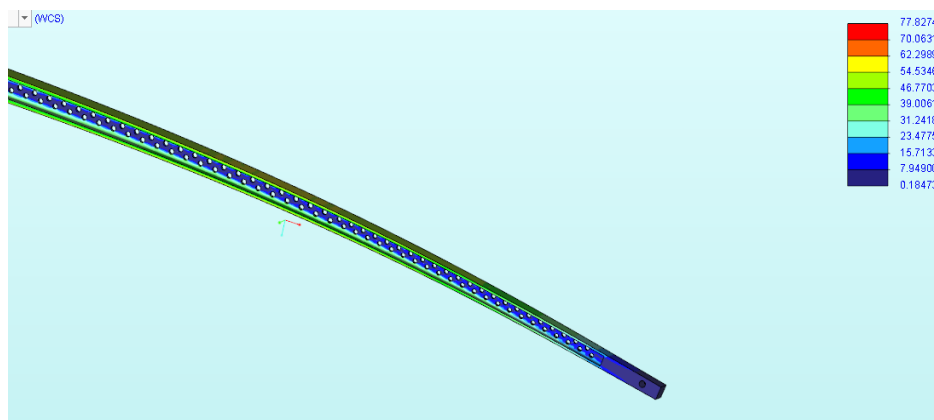


Fig. I.4: AL 7075-T6 Modified I-beam with 1/16 in radius fillet and original continuous hole pattern (no bridge) with 95 lbf force acting on end.

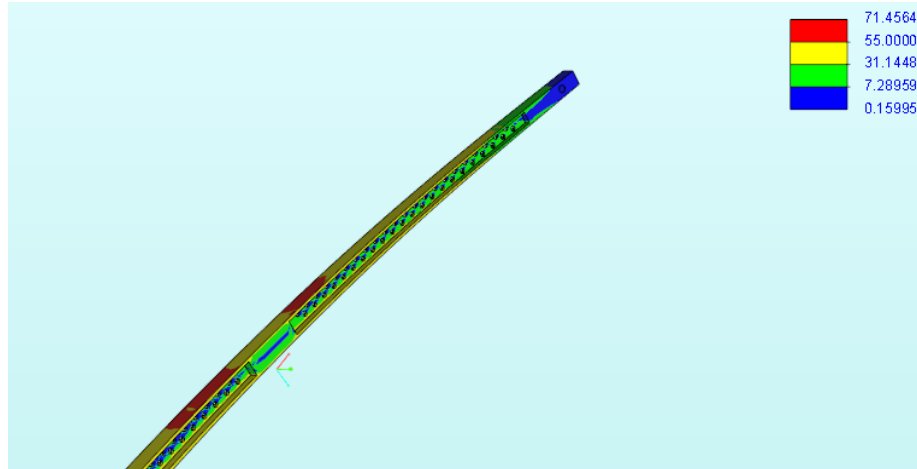


Fig. I.5: AL 7075-T6 Modified I-beam with central bridge and separated hole patterns (no fillet) with 95 lbf force acting on end.

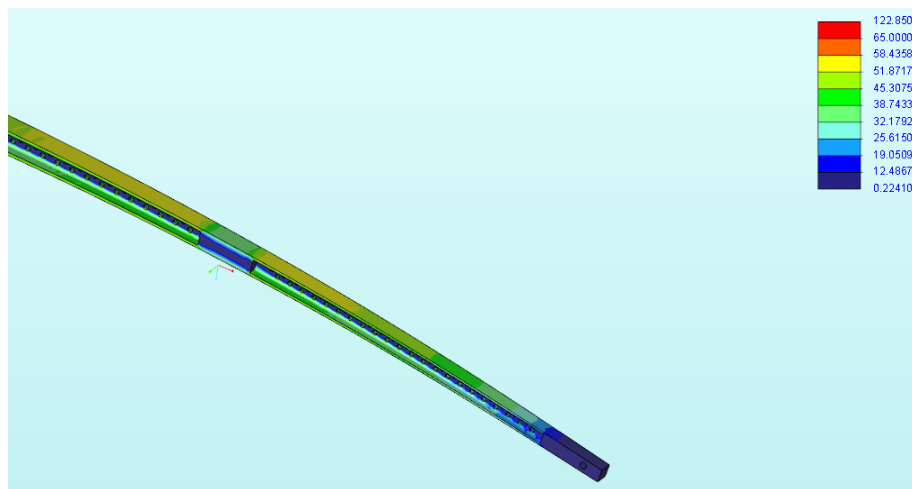


Fig. I.6: AL 7075 Final modified I-beam with central bridge, separated hole patterns and 1/16 fillet with 95 lbf force acting on end.

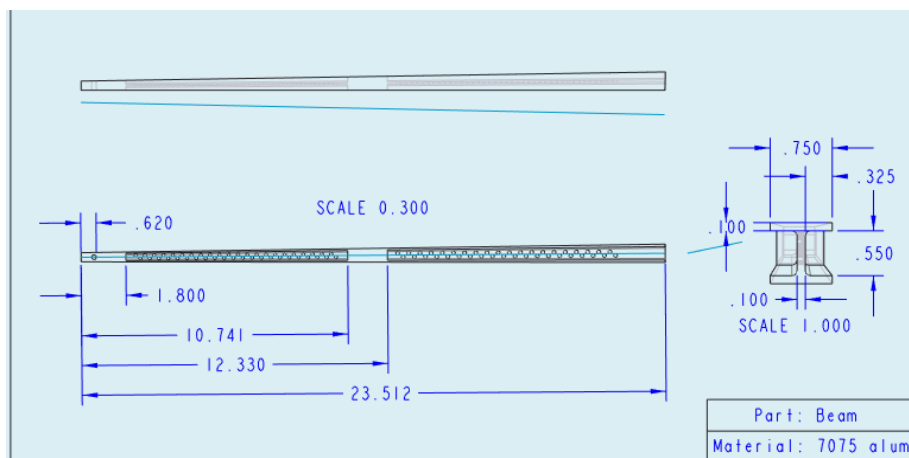


Fig. I.7: Engineering drawing of the final beam design.

II. Bulkheads

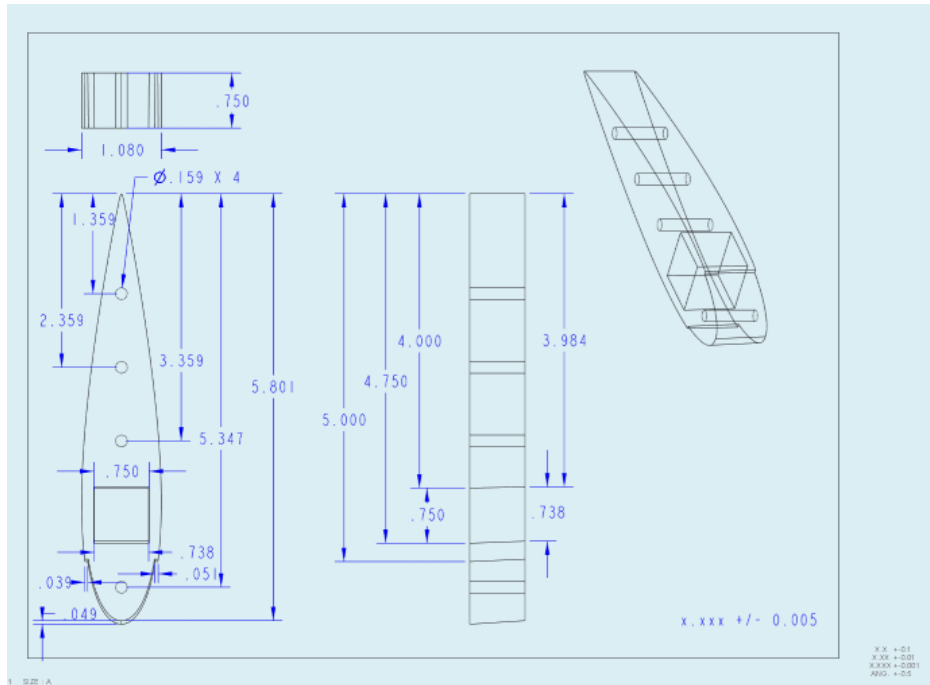


Fig. II.1: Engineering drawing of bulkhead #1.

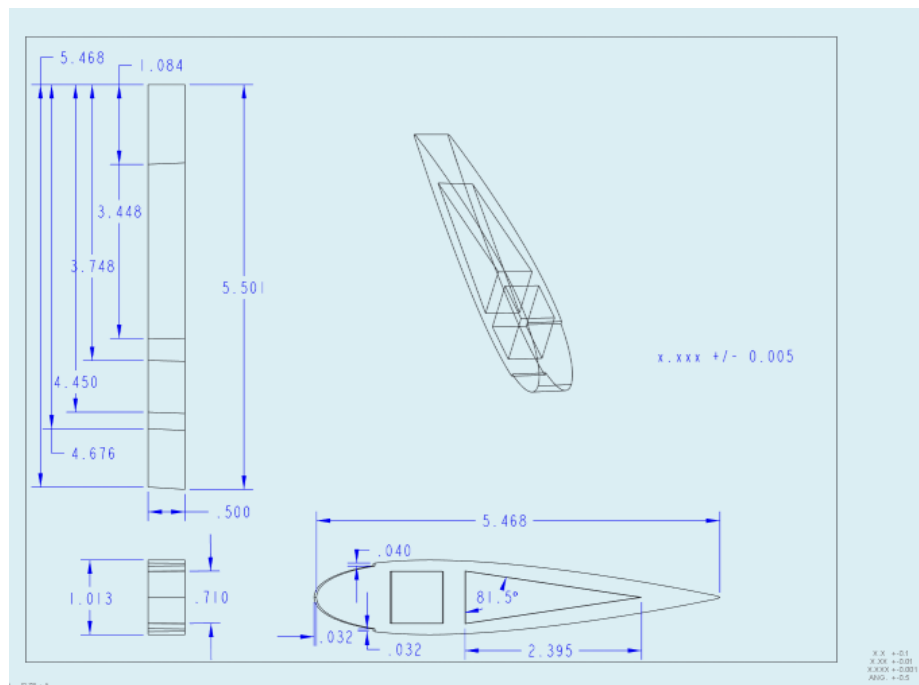


Fig. II.2: Engineering drawing of bulkhead #2

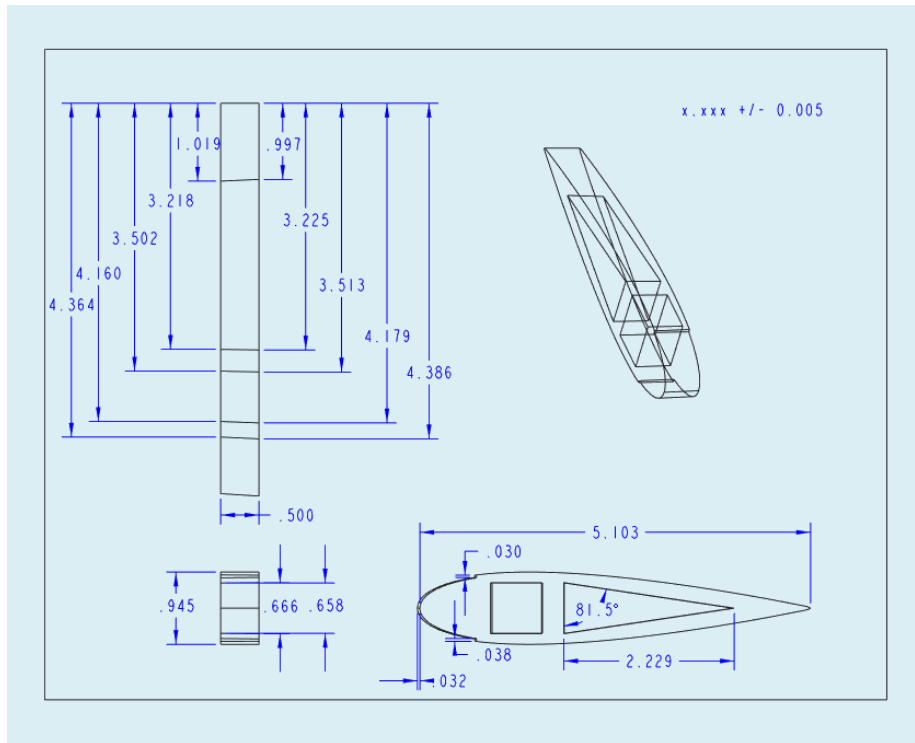


Fig. II.3: Engineering drawing of bulkhead #3

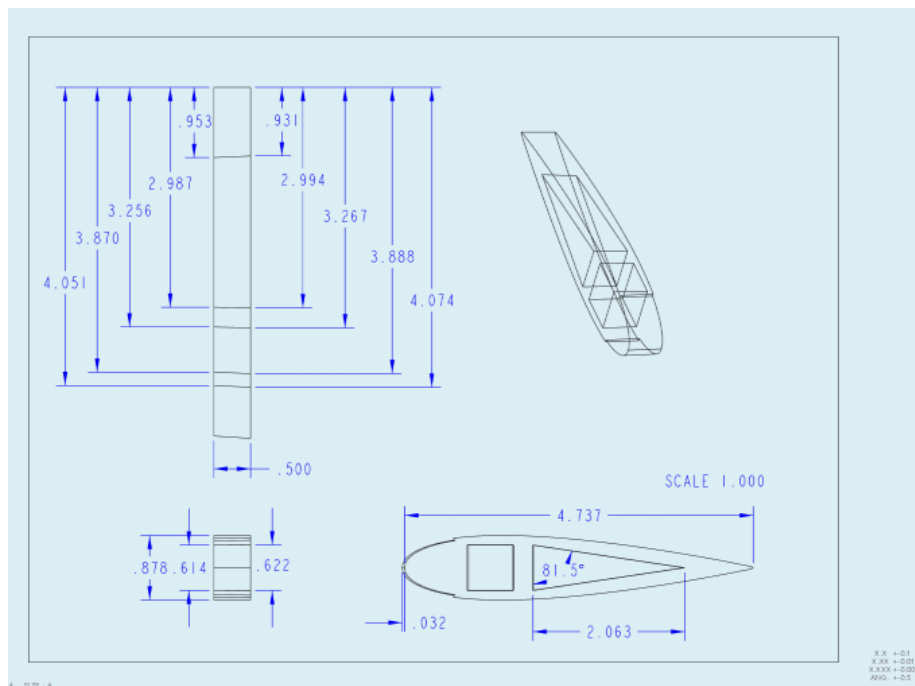


Fig. II.4: Engineering drawing of bulkhead #4

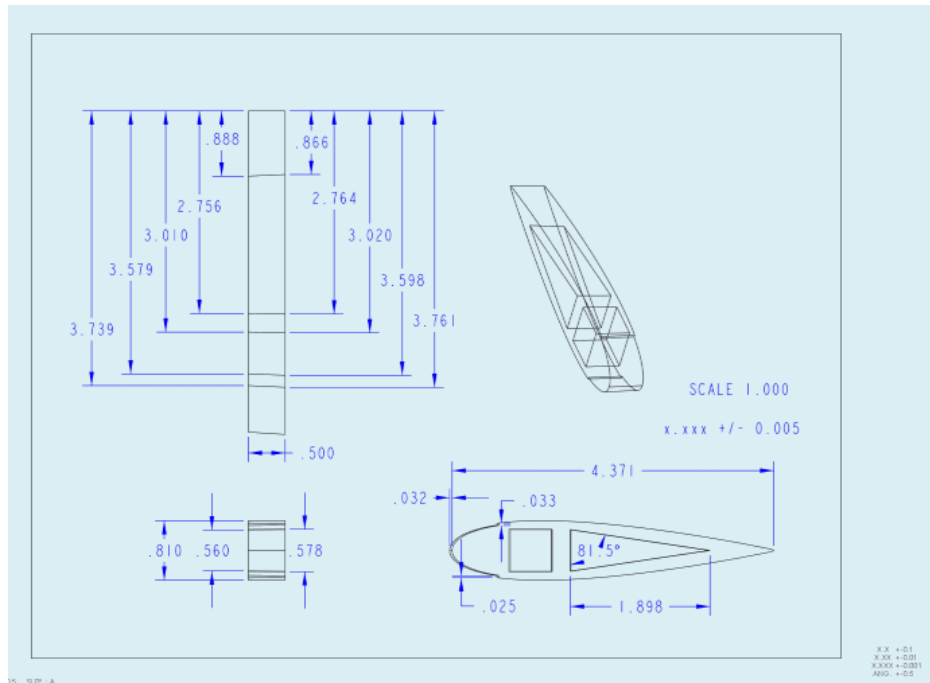


Fig. II.5: Engineering drawing of bulkhead #5

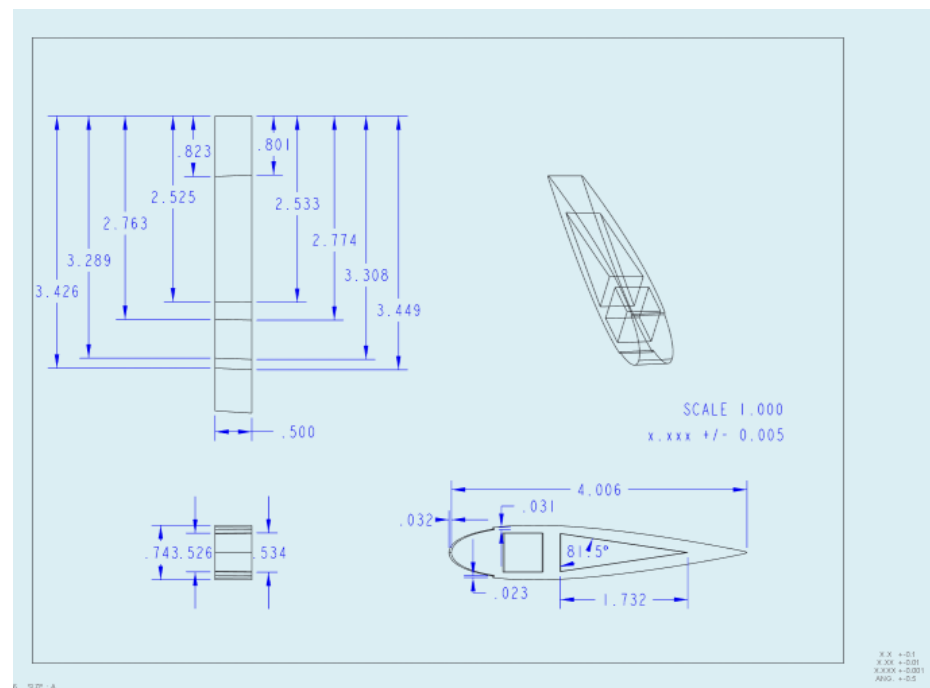


Fig. II.6: Engineering drawing of bulkhead #6

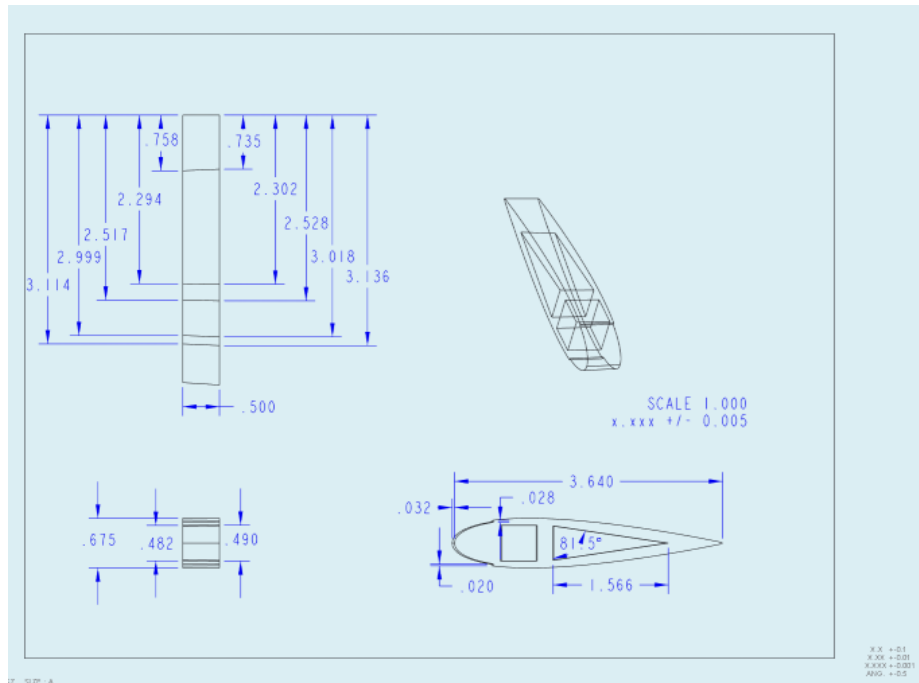


Fig. II.7: Engineering drawing of bulkhead #7

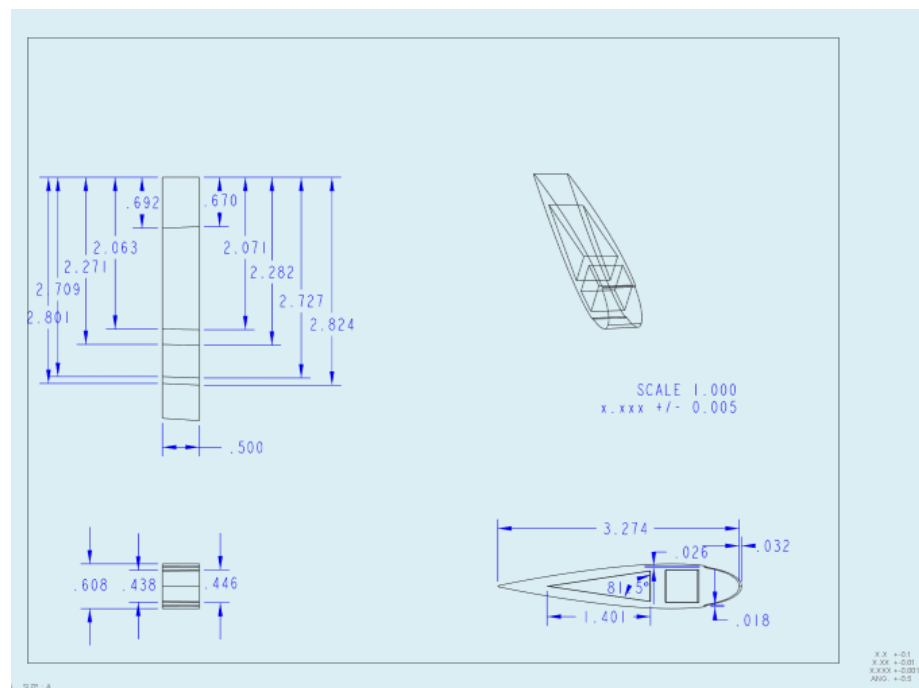


Fig. II.8: Engineering drawing of bulkhead #8

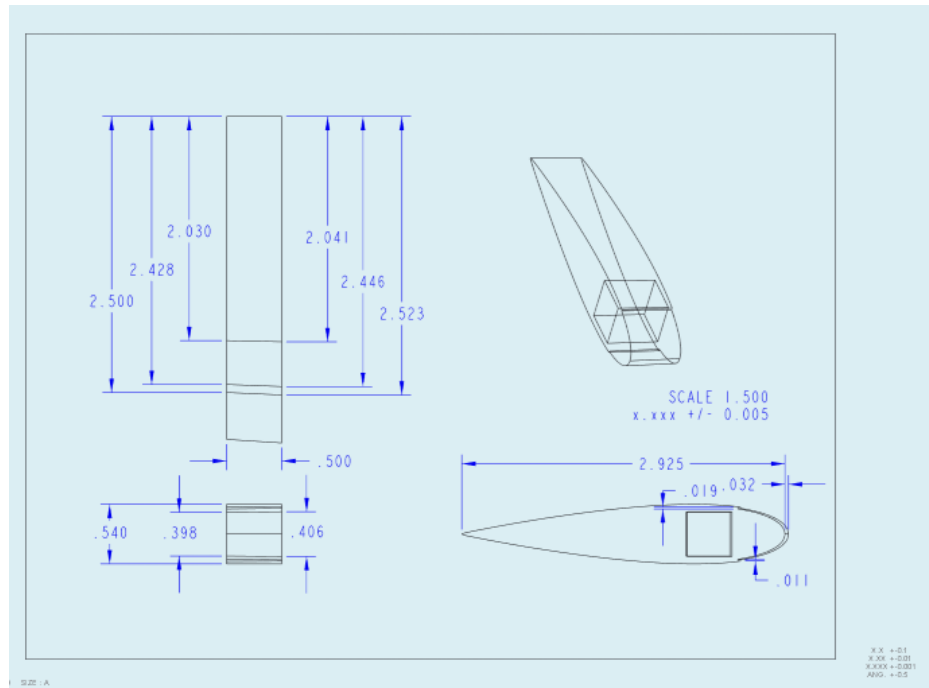


Fig. II.9: Engineering drawing of bulkhead #9

Number of Bulkheads	Beam Load Factor (BLF)
9	-1.315109
8	0.9227785
7	-1.817586
6	0.7221506
5	0.8020391

Table II.1: The corresponding BLF for a decreasing number of bulkheads with downwards load 90 lbf and material AL6061.

III. Airfoil Surface

Diagram	Material	Thickness (in)	Yield Stress (ksi)	Approx. Stress on Skin (ksi)
III.1	7075-T6 (top) 2024-T3 (bottom)	0.05 0.05	61 (McMaster)	10
III.2	7075-T6 (top) 2024-T3 (bottom)	0.04 0.04	61 (McMaster)	13
III.3	7075-T6 (top) 2024-T3 (bottom)	0.032 0.02	61 (McMaster)	15
III.4	2024-T3 (top) 2024-T3 (bottom)	0.02 0.02	41 (McMaster)	22

Table A2 - CREO Static Stress Analysis for 7075-T6 and 2024-T3

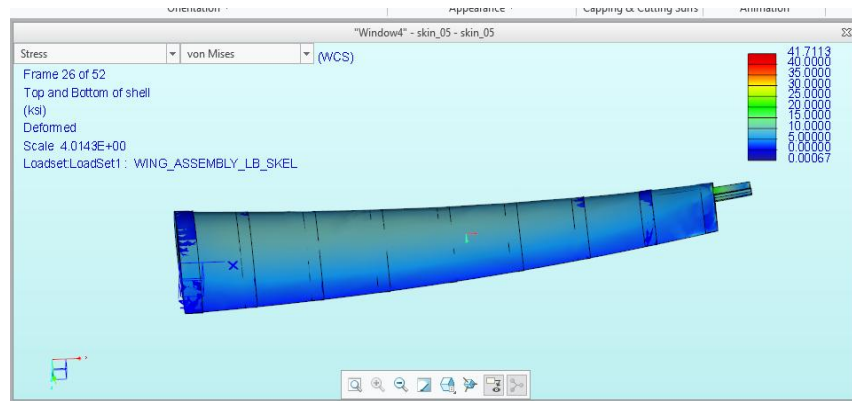


Fig. III.1: Simulated skin stresses for thicknesses of 0.05” on top and bottom.

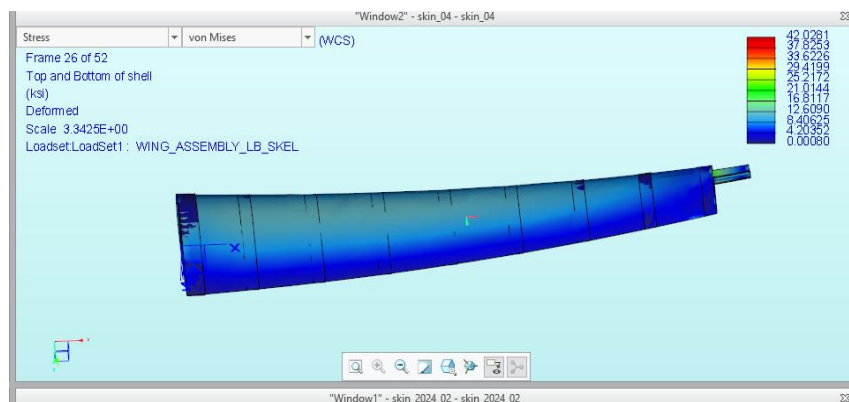


Fig. III.2: Simulate skin stresses for thicknesses of 0.04” on top and bottom.

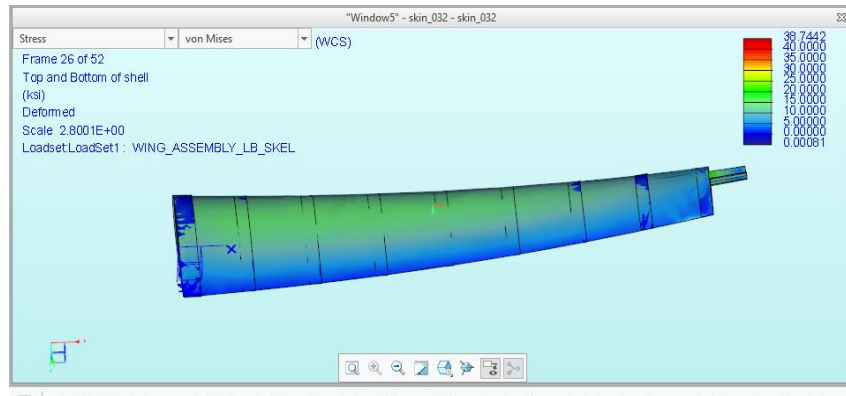


Fig. III.3: Simulated skin stresses for thicknesses of 0.032” on top and 0.02” on bottom.

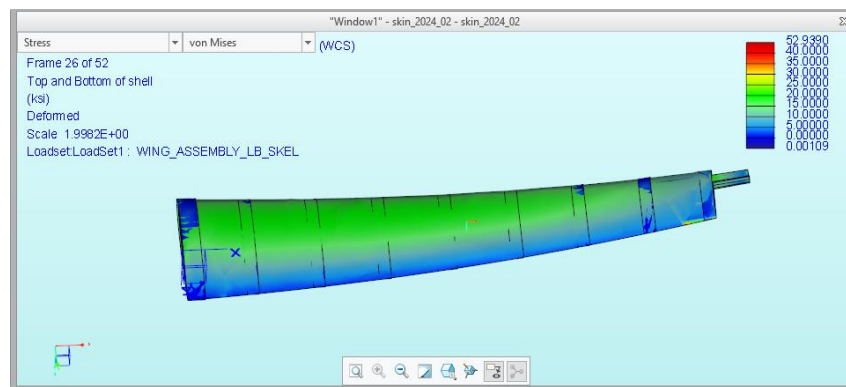


Fig. III.4: Simulated skin stresses for thicknesses of 0.02” on top and bottom (using AL 2024 on top, as AL 7075 does not come thinner than 0.032”).

IV. Component Fasteners

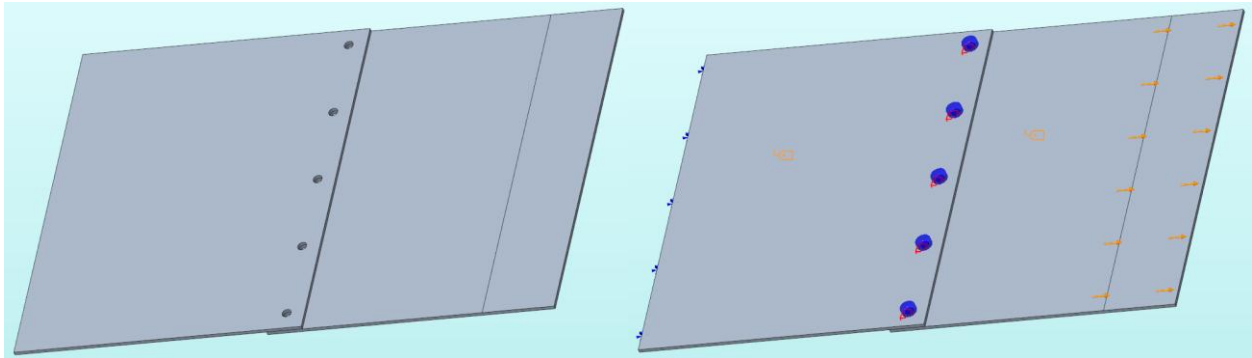


Fig. IV.1: (left) The model used to test the fastener strength, shown here using the final design of 5 rivets. (right) The testing model fully prepared for simulation, with the leftmost surface fixed, CREO fasteners through each hole, and the test load pulling on a portion of the right plate.

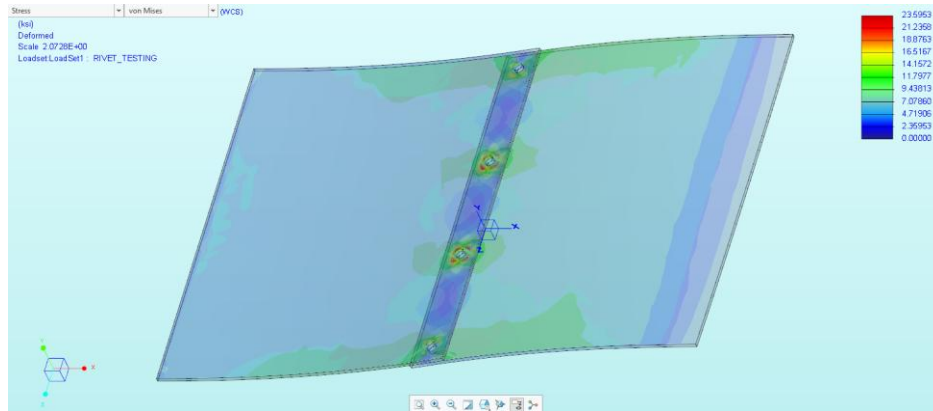


Fig. IV.2: Results of testing the fastener apparatus with 4 rivets, displaying a maximum predicted stress of about 23.59 ksi around the holes.

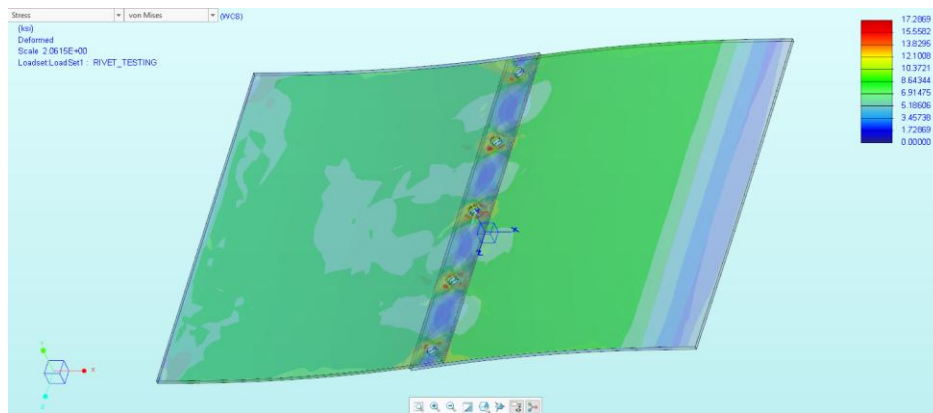


Fig. IV.3: Results of testing the fastener apparatus with 5 rivets, displaying a maximum predicted stress of about 17.29 ksi around the holes.

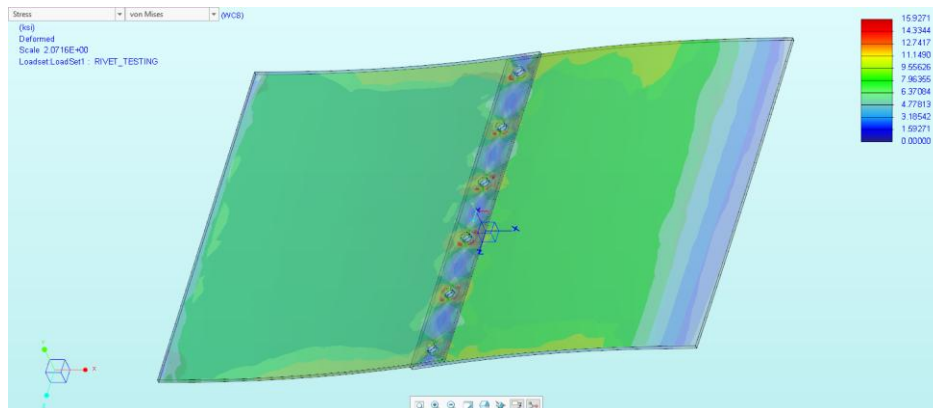


Fig. IV.4: Results of testing the fastener apparatus with 6 rivets, displaying a maximum predicted stress of about 15.93 ksi around the holes.

V. Additional Operational Analyses

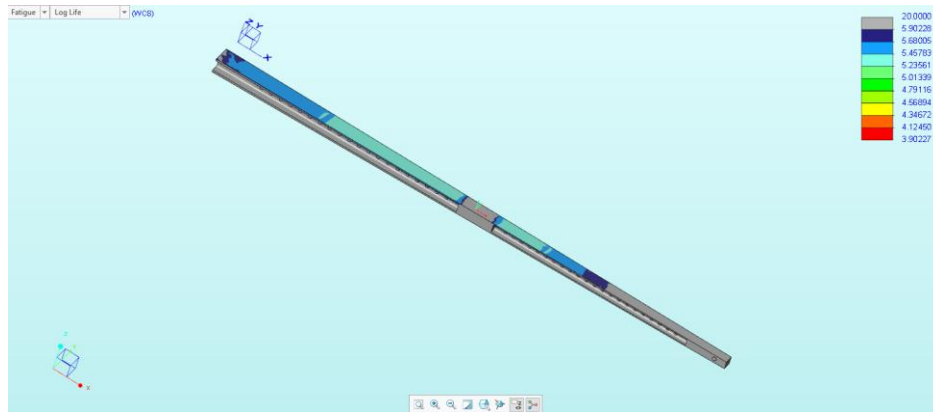


Fig. V.1: Results of the fatigue analysis on the wing beam model, showing predicted failure along the top flange after 5 cycles.

VI. Testing

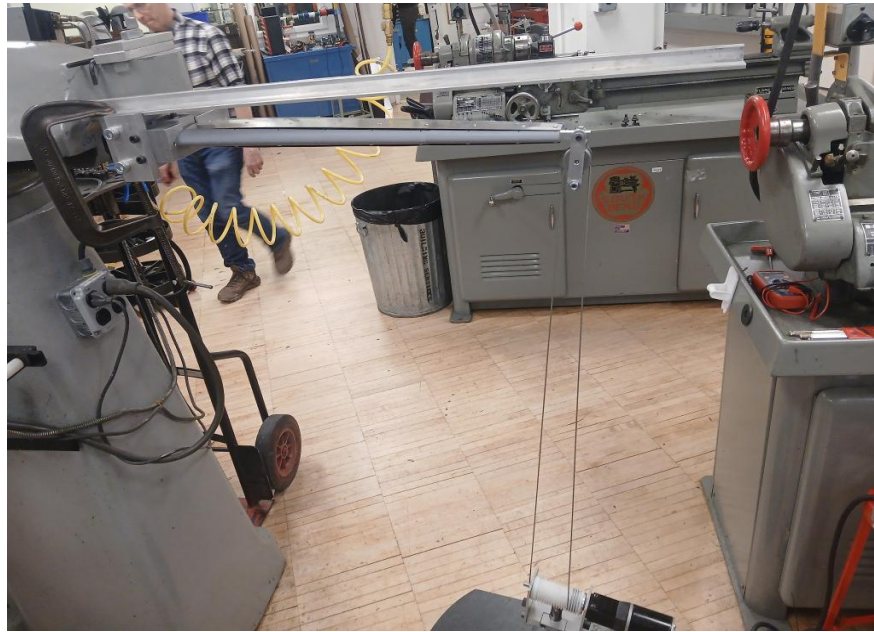


Fig. VI.1: The wing secured to the testing apparatus, prior to the application of any load.

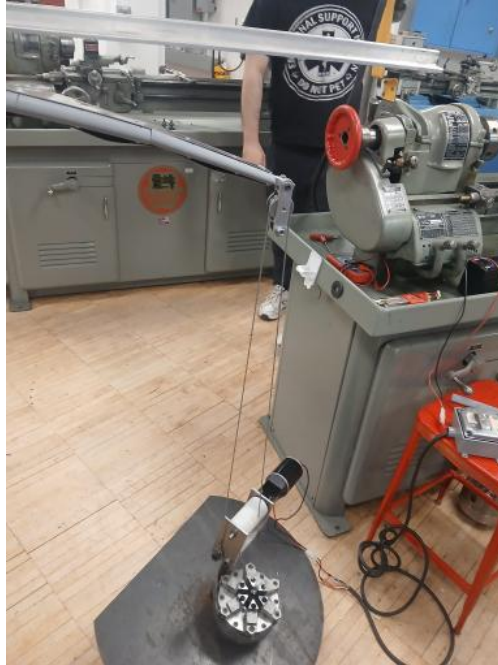


Fig. VI.2: The wing with the specified 95.5-lb load applied during testing, displaying some deflection.

VII. Results



Fig. VII.1: Buckling of skin and fracturing of leading edge.



Fig. VII.2: Cracks in the leading edge near the rivets.

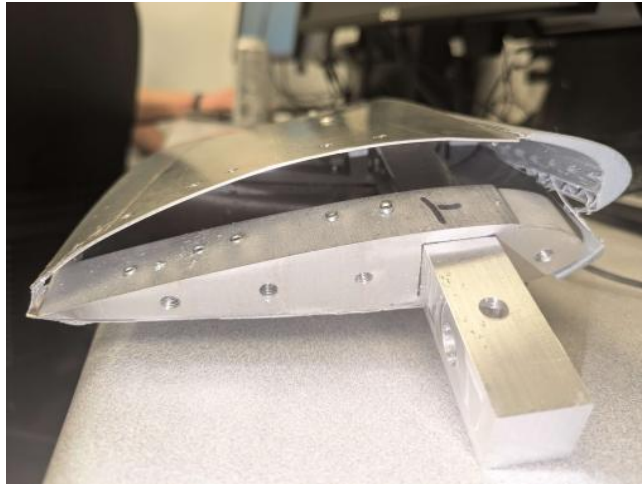


Fig. VII.3: Popped rivets, stretched and torn tape on trailing edge, deformed beam.



Fig. VII.4: Separation of the skin from the bulkheads



Fig. VII.5: Delamination of the bulkheads, buckling of the skin, and fracturing of the leading edge.

Deflection vs. Weight

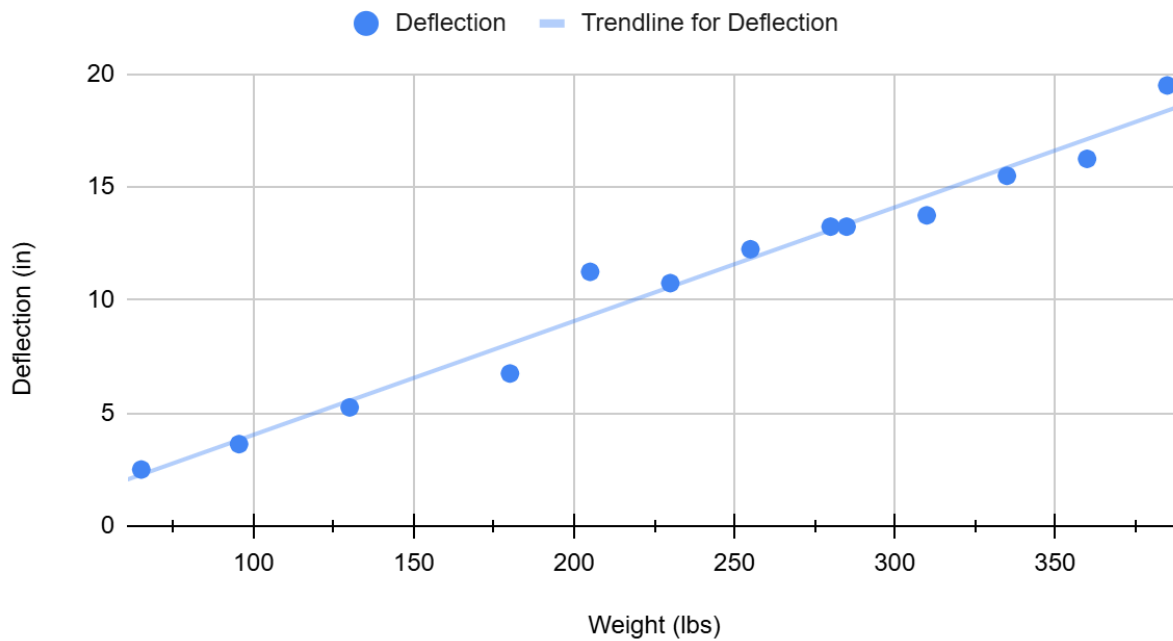


Fig. VII.6: A graph showing Deflection vs Weight.

Weight (lbs)	Deflection (in)	Change in Deflection (in)
65	2.5	
95.5	3.625	1.125
130	5.25	1.625
155	*not recorded*	
180	6.75	1.5
205	11.25	4.5
230	10.75	-0.5
255	12.25	1.5
280	13.25	1
285	13.25	0
310	13.75	0.5
335	15.5	1.75
360	16.25	0.75
385	19.5	3.25

Table VII.1: Corresponding deflection and change in deflection for each trial of a specified weight.